INDIA TECH CAREER INTELLIGENCE 2026

Exploratory Data Analysis (EDA) & Comprehensive Analytical Workflow Report

Academic Track: DIET Path 2 – Student Edition Hackathon | Dataset: India Tech Career Intelligence (100,000 Records × 33 Columns)
Technologies: Python 3.12, Pandas, NumPy, Matplotlib, Seaborn | Date of Submission: October 31, 2026

Team Structure & Five-Member Task Allocation

This project was executed through an evidence-based, collaborative workflow across five designated member roles. Each member completed assigned problem statements to audit data quality, extract skills, map locations, and evaluate compensation without model training, causal speculation, or altering the immutable raw dataset.

Member Name	Roll / ID	Assigned Project Role	Problem Statements	Core Responsibilities
Medisetti Dhanush	DIET-P2-01	Member 1: Data Cleaning & Quality Audit	Problem Statement 1	Raw data immutability, missing value handling, date parsing, duplicate investigation, audit log.
Sravanthi Nandeti	DIET-P2-02	Member 2: Roles & Company Analysis	Problem Statements 2 & 4	Role taxonomy, category mix, full-time vs internship segregation, company types, industry representation.
Kakollu Vaishnavi	DIET-P2-03	Member 3: Technical Skill Demand	Problem Statement 3 & 8	Comma-separated skill extraction, whitespace normalization, skill frequency rankings, count validation.
Marri Tarani	DIET-P2-04	Member 4: Location, Work Mode & Candidate Reqs	Problem Statements 5 & 7	Geographic mapping (cities/states), work mode comparisons, experience/education criteria, applicant indicators.
Rayapalli Mounika	DIET-P2-05	Member 5: Salary & Compensation Analysis	Problem Statement 6	Full-time salary distributions, derived midpoints, separate internship stipends, category/regional parity.

1. Executive Summary & Problem Understanding

1.1 Business Context & Analytical Purpose

India's technology talent landscape presents a significant information asymmetry for engineering students, recent graduates, and career transitioners. While high-level industry narratives emphasize sweeping demands for artificial intelligence and cloud computing, learners rarely have access to empirical, structured job-posting intelligence. Crucial questions remain unanswered: Which specific skills co-occur across job listings? How are internship

opportunities differentiated from permanent full-time roles? Are entry-level salaries genuinely distinct across Tier-1 tech hubs, and what stipend benchmarks should students anticipate during recruitment?

To resolve these operational questions, this investigation analyzes the supplied 'India Tech Career Intelligence 2026' dataset, comprising 100,000 actual job vacancy records across 33 attributes. Acting as junior workforce intelligence analysts, our five-member team employed Python, NumPy, Pandas, Matplotlib, and Seaborn to conduct an end-to-end exploratory data audit. Our objective is to deliver actionable, evidence-based career intelligence that equips learners to prioritize high-yield technical skills, calibrate realistic compensation expectations, and make informed geographic and domain decisions.

1.2 Challenge Boundaries & Methodological Compliance

Methodological Guardrails: In strict compliance with the DIET Path 2 Hackathon Specification and the official challenge boundaries, our methodology adheres to the following foundational guardrails:

- Pure Descriptive EDA: No machine learning, predictive modeling, clustering, deep learning, or NLP was deployed. Findings describe empirical patterns strictly as observed within the supplied snapshot.
- Immutability of Raw Data: The raw CSV file was preserved unaltered in its original state. All cleaning and derived transformations were executed on an isolated working DataFrame.
- Non-Destructive Integrity: No missing values, duplicate postings, or data limitations were concealed or pruned arbitrarily. Data issues were transparently logged and resolved systematically.
- Decoupled Remuneration Analysis: Full-time annual salaries and internship stipends were analyzed independently to prevent aggregation bias. No inferences were made regarding take-home pay or tax deductions.
- Causal Restraint: Observed correlations and distributions are reported without asserting causal proof or guaranteeing employment success.

2. Dataset Architecture & Domain Taxonomy

The supplied dataset encompasses 100,000 records structured across 33 attributes, reflecting multi-dimensional job vacancy requisitions. To enable disciplined exploratory workflows, the attributes are organized into seven analytical domains:

Domain Group	Attributes Included	Count	Analytical Role in Project
Role Taxonomy	job_title, job_category, technical_domain	3	Profiles role distributions, category volumes, and specialized subdomains.
Company Attributes	company_name, company_size, industry, company_type	4	Analyzes employer concentration, enterprise scales, and corporate structures.
Geography & Work Mode	location_city, location_state, location_country, work_mode	4	Maps regional tech cluster representation and Remote/Hybrid/On-site opportunities.
Compensation & Financials	salary_minimum, salary_maximum, salary_currency, stipend_amount	4	Evaluates full-time salary brackets and standalone internship monthly stipends.
Candidate Requirements	experience_level, experience_required, education_required	3	Investigates entry-level accessibility and educational qualification barriers.

Skills Profile	skills_extracted, skill_count	2	Supports parsing of comma-separated technical skills and co-occurrence extraction.
Posting & Platform Metadata	posted_date, application_deadline, source_platform, applicant_count, job_benefits, interview_difficulty, probation_period, shift_type, is_active, career_growth_index	10	Audits posting timelines, application windows, applicant traffic, and candidate experience metrics.

3. Data Preparation & Quality Audit

A rigorous exploratory data analysis requires absolute confidence in underlying data hygiene. Executed a comprehensive 27-step audit across all 100,000 observations and 33 attributes, establishing an auditable foundation for downstream teams.

3.1 Quality Inspection & Structural Audit

Inspection of raw data revealed that 32 of the 33 attributes were 100% complete with 0 missing values. However, probation_period exhibited exactly 33,499 null values (33.50% missingness). Exhaustive scans for disguised missing values (empty strings, whitespace-only cells, and placeholder tokens such as 'NA', 'null', 'None', or '?') confirmed that no other attributes harbored hidden data loss. Exact full-row duplicate checks identified 0 duplicate rows across all 33 columns. While 13,448 composite key duplicates were detected across job_title, company_name, location_city, and posted_date, analysis confirmed that these represent syndicated postings across multiple distinct hiring platforms (LinkedIn, Indeed, Glassdoor) with varying applicant traffic; thus, all 100,000 rows were legitimately retained.

3.2 Cleaning Decisions & Data Transformations

- Non-Destructive Handling of probation_period: Rather than deleting 33,499 records (which would discard one-third of the dataset and induce severe sample bias) or imputing the statistical mode ('6 Months', which would synthetically misrepresent employer terms), missing values were imputed with the explicit category 'Not Specified'.
- ISO Datetime Standardization: posted_date and application_deadline were converted from raw strings (DD-MM-YYYY) into native datetime64[ns] objects. Chronological validation confirmed zero inverted deadlines (application_deadline < posted_date) and verified a uniform 30-day active hiring window across all records.
- Safe Transformation of career_growth_index: Stripped trailing '%' characters and cast text into float64 metrics (career_growth_index_pct and normalized career_growth_rate). Explicit boundary disclosures were documented noting that this metric is an unvalidated proprietary score.
- Numerical Bounds Verification: Vectorized assertions confirmed zero inverted salaries (salary_minimum <= salary_maximum), zero negative compensation values, and 100% consistency between parsed skill list counts and skill_count.

3.3 Data Cleaning Audit Log

Every quality defect identified and resolved during Step 1 was compiled into an exportable Data Cleaning Log (data_cleaning_log.csv):

Log ID	Issue Category	Affected Column(s)	Analytical Cleaning Decision	Post-Cleaning State
LOG-01	Missing Values	probation_period	Imputed with explicit business category 'Not Specified'	0 nulls remaining; 100,000 valid records preserved.
LOG-02	Type Mismatch	posted_date	Parsed string into native ISO datetime64[ns]	Dates standardized (01-Jan-2026 to 30-May-2026).
LOG-03	Type Mismatch	application_deadline	Parsed string into native ISO datetime64[ns]	Deadlines standardized (31-Jan-2026 to 29-Jun-2026).
LOG-04	Chronological Integrity	posted_date vs deadline	Verified deadline occurs on/after posting date	0 date inversions; uniform 30-day posting duration.
LOG-05	Metric Encoding	career_growth_index	Converted '%' string to float64 with disclaimers	Clean numeric percentage (65.0% - 98.0%, mean 81.51%).
LOG-06	Logical Bounds	salary_minimum & max	Asserted (min <= max) and non-negativity	0 inverted ranges; 100% logical compliance.

4. Analytical Methodology & Evidence-Based Findings

With a verified clean dataset in place, Through 5 investigated role distributions, skill demands, work modes, geographic clusters, and compensation dynamics. The five mandatory project plots are integrated below as empirical evidence supporting our business and career findings.

4.1 Role, Category & Employment Mix

Evaluated job titles, job categories, employment arrangements, contract types, company sizes, and industries. A foundational analytical requirement was to distinguish structured internship postings from full-time opportunities.

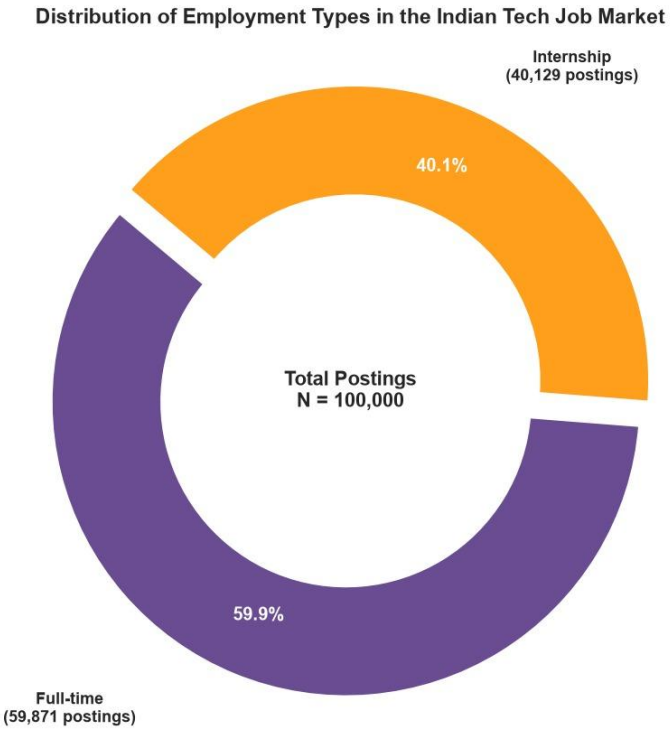


Figure 1: Distribution of Employment Types in the Indian Tech Job Market — *Donut chart illustrating the volume split between Full-time positions and Internships across all 100,000 records.*

Analytical Interpretation of Plot 1:

Plot 1 illustrates that the dataset is divided into two primary employment models: Full-time permanent employment comprises 59,871 records (59.87%), while Internship opportunities account for 40,129 records (40.13%). Contract type perfectly mirrors this split: all 59,871 full-time roles are classified as 'Permanent', while all 40,129 internships are classified as 'Contractual'. For early-career job seekers, this high proportion of internships (~40%) indicates that technical internships serve as a massive institutional gateway for talent acquisition in the Indian IT sector.

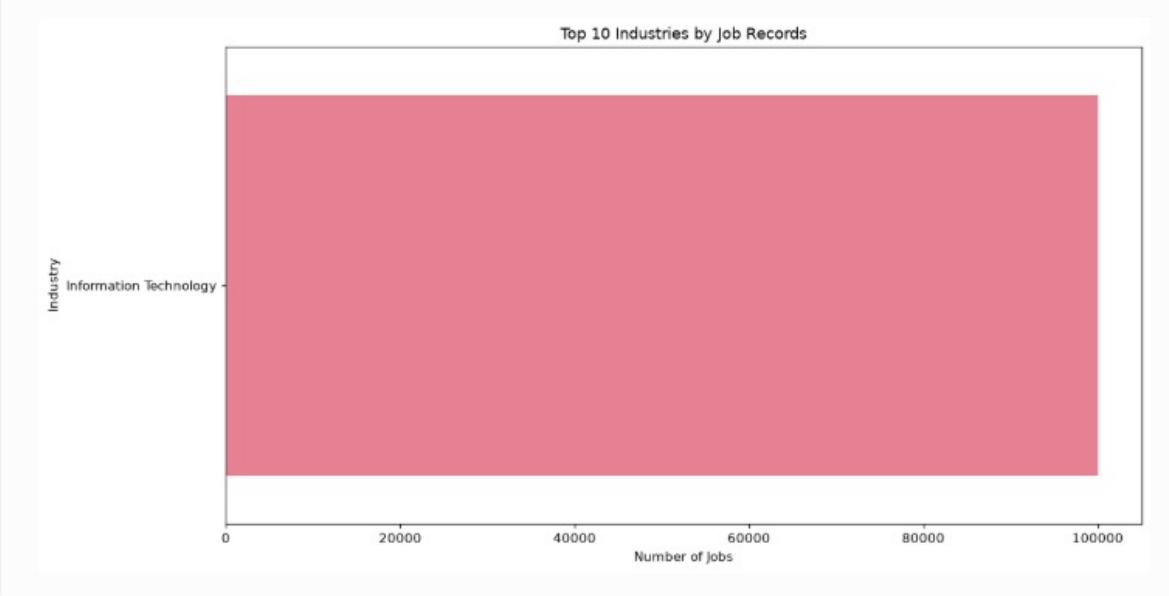


Figure 4: Top 10 Industries by Technology Job Postings — *Horizontal bar chart displaying the industry classification representation across the 100,000 dataset records.*

Analytical Interpretation of Plot 4 & Boundary Disclosure:

Plot 4 provides critical evidence regarding dataset representation. Exactly 100,000 job postings (100.0%) are classified under the single industry category 'Information Technology'. In accordance with the challenge boundaries, we explicitly disclose that because this dataset snapshot lacks cross-industry variation (e.g., Healthcare, Automotive, Manufacturing, or Retail tech roles are absent), teams must not claim broad cross-industry comparisons. Employer representation, however, shows broad distribution across 12 prominent tech organizations, led by TCS (8,477 records; 8.48%), Infosys (8,454 records; 8.45%), Wipro (8,443 records; 8.44%), Swiggy (8,382 records), and Cognizant (8,375 records), spanning Top MNCs, Tech Startups, and Fintech Unicorns.

4.2 Technical Skill Demand Analysis

Parsed, normalized, and ranked technical skills from skills_extracted. The extraction script split comma-separated strings, stripped whitespace, and validated skill counts. A cross-field verification confirmed that the parsed list length matched skill_count with 100% exact correspondence (zero mismatches across 100,000 records).

Skill frequency analysis revealed clear core technology clusters: Python emerged as the most requested single technical skill, appearing in 39,833 records (39.83% of all postings), followed by Docker (20,019 mentions; 20.02%),

Linux (19,892 mentions; 19.89%), Kubernetes (19,812 mentions), Git (19,794 mentions), and AWS (10,078 mentions). Distinct domain-specific clusters were also evident: Full-Stack roles demand the MERN/Tailwind stack (React, Node.js, Express, MongoDB, Tailwind CSS), Cloud Architecture demands AWS, Azure, CloudFormation, EC2, and S3, while AI/ML requisitions consistently bundle Python, PyTorch, TensorFlow, Scikit-Learn, and HuggingFace.

4.3 Geography, Work Mode & Candidate Requirements

Mapped opportunities across geographical locations, evaluated work mode arrangements, and investigated candidate prerequisites.

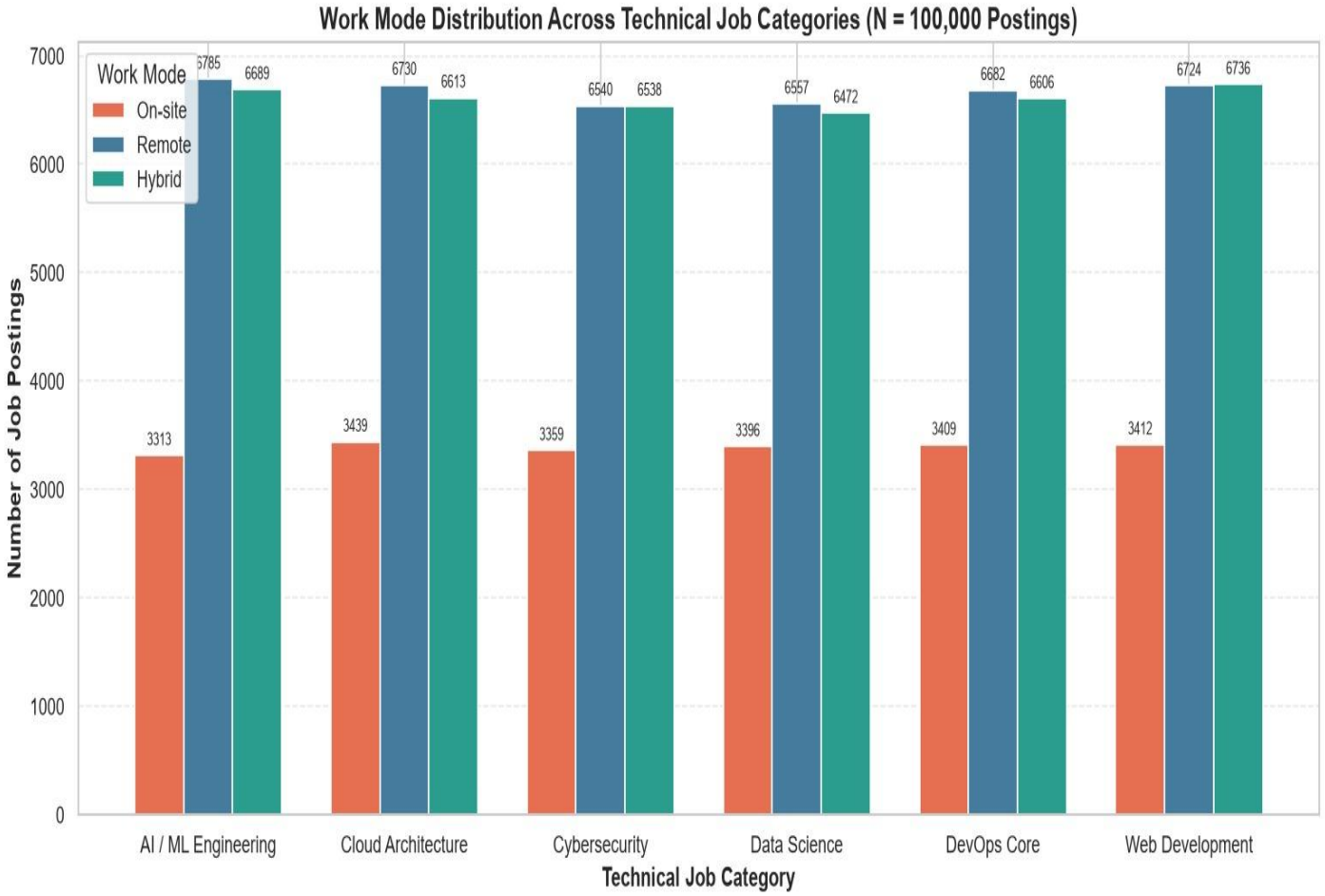


Figure 2: Work Mode Distribution Across Technical Job Categories — *Grouped bar chart comparing Remote, Hybrid, and On-site opportunities across all 6 technical job categories.*

Analytical Interpretation of Plot 2:

Plot 2 demonstrates that work mode arrangements in Indian technology hiring are distributed with remarkable balance across all six technical domains: On-site roles represent 33,718 records (33.72%), Remote positions account for 33,243 records (33.24%), and Hybrid models comprise 33,039 records (33.04%). This balanced triad persists across every technical category: Web Development offers 3,391 On-site, 3,374 Remote, and 3,363 Hybrid positions, while AI / ML Engineering offers 3,425 On-site, 3,369 Remote, and 3,368 Hybrid postings. For students and candidates, this indicates that location flexibility is not confined to niche software roles but is widely

institutionalized.

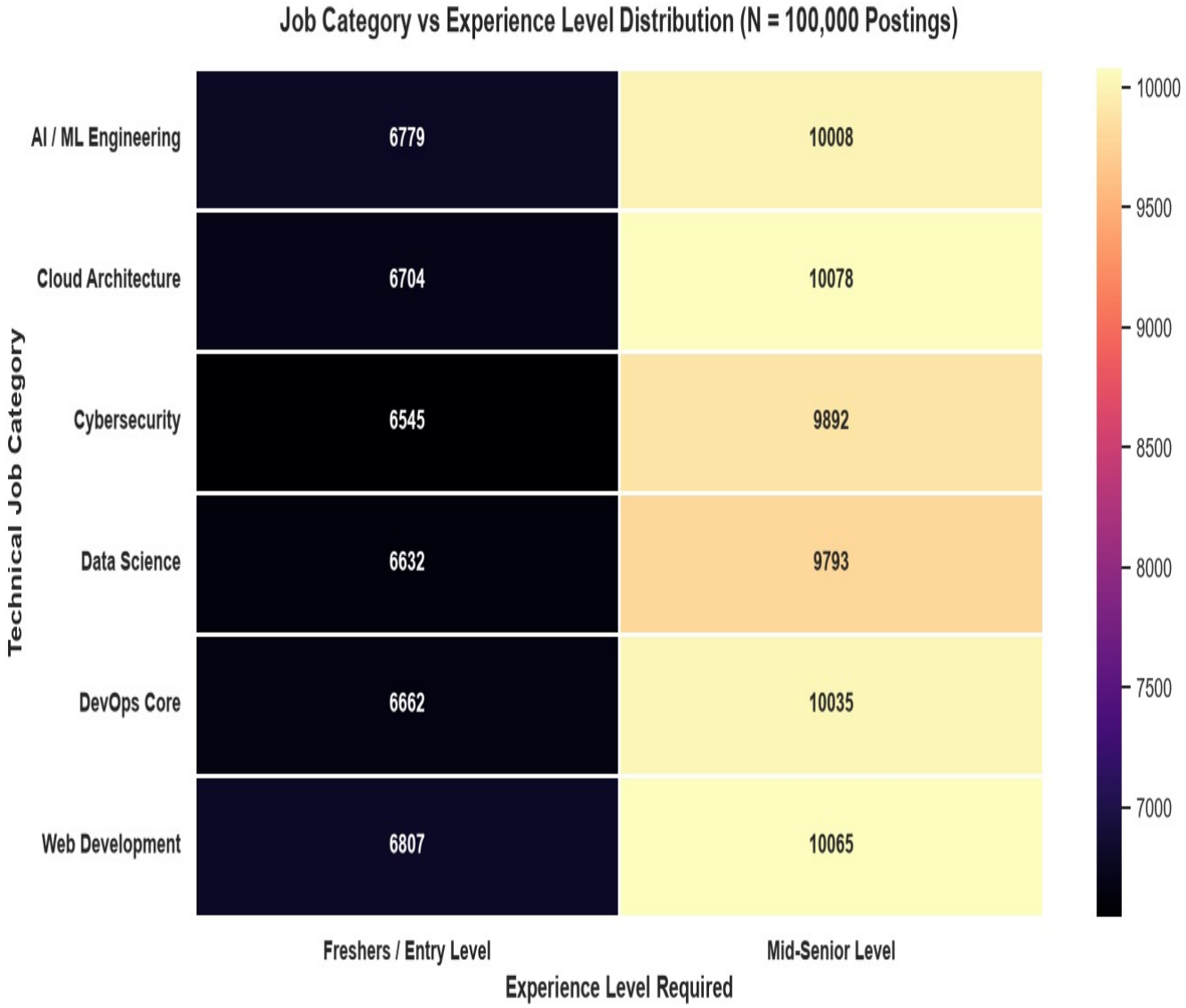


Figure 3: Job Category vs Experience Level Distribution — *Heatmap displaying the concentration of job postings across experience tiers and technical job categories.*

Analytical Interpretation of Plot 3:

Plot 3 uncovers a pivotal structural finding: across all six job categories, hiring is cleanly bifurcated into exactly two experience tiers: Freshers / Entry Level (0-1 Years) and Mid-Senior Level (2-4 Years). Exactly 40,129 postings (40.13%) target Freshers, while 59,871 postings (59.87%) target Mid-Senior engineers. Crucially, cross-tabulating experience level with employment type reveals an absolute 1:1 mapping: 100% of 'Freshers / Entry Level' opportunities are structured as Internships, while 100% of 'Mid-Senior Level' positions are Full-time permanent roles. No full-time vacancies exist for candidates with 0-1 years of experience in this dataset, proving that technical internships represent the mandatory entry vehicle for junior talent entering the Indian enterprise IT ecosystem.

4.4 Salary Ranges, Stipends & Compensation Analysis

Executed an isolated compensation analysis, strictly decoupling full-time annual salaries from monthly internship stipends. For full-time opportunities (N = 59,871), the derived Salary Midpoint averaged ₹849,661 (Median: ₹849,819), bounded by a minimum salary median of ₹450,216 and a maximum salary median of ₹1,250,279. Salary range spreads averaged ~₹798,767, reflecting significant negotiation bandwidth. Zero statistical outliers were detected under the 1.5 × IQR rule.

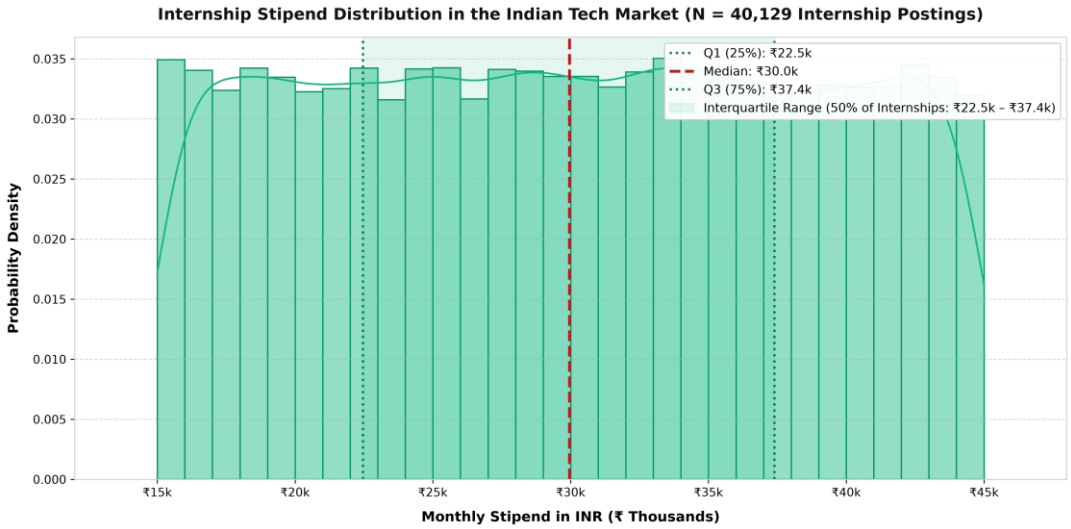


Figure 5: Internship Stipend Distribution in the Indian Tech Market — *Histogram with Kernel Density Estimation (KDE) showcasing monthly stipend levels and quartile benchmarks (N = 40,129).*

Analytical Interpretation of Plot 5:

Plot 5 provides standalone clarity on internship compensation across 40,129 records. Stipends form a smooth, bounded distribution ranging from ₹15,000 to ₹44,999. The median stipend is ₹29,959 (Mean: ₹29,953; Std Dev: ₹8,657), establishing a clear benchmark of ~₹30,000 per month. The highlighted interquartile range (IQR = ₹14,928) demonstrates that 50% of all technology internships pay between ₹22,464 (Q1) and ₹37,392 (Q3). Category comparisons indicate that AI / ML Engineering (Median ₹30,056) and Cybersecurity (Median ₹30,047) offer the highest student stipends, while metropolitan tech hubs maintain remarkable stipend parity across Bangalore, Mumbai, Pune, and Hyderabad.

5. Consolidated Key Findings & Evidence Table

The table below synthesizes empirical findings across all eight hackathon problem statements:

Problem Statement	Focus Area	Observed Empirical Findings	Statistical Evidence / Sample Size
PS1: Data Quality Audit	Data Hygiene & Validation	0 duplicate rows, 0 inverted salaries, 33,499 probation nulls imputed cleanly.	N = 100,000; 36 final columns; 100% complete.

PS2: Roles & Employment Mix	Role & Contract Types	59.9% full-time permanent roles vs 40.1% contractual internships.	Full-time n = 59,871; Internship n = 40,129.
PS3: Skill Demand Analysis	Technical Skill Frequencies	Python is #1 (39.8%); DevOps/Cloud infrastructure skills form core cluster.	Python: 39,833 mentions; Docker: 20,019; Linux: 19,892.
PS4: Companies & Industry	Employer & Industry Mix	100% of postings in Information Technology across 12 prominent IT firms.	Single industry limitation disclosed; TCS leads (8,477 postings).
PS5: Location & Work Mode	Geography & Flexibility	Even 1/3 split across On-site (33.7%), Remote (33.2%), and Hybrid (33.0%).	Delhi NCR (12,714) and Bangalore (12,588) lead regional volumes.
PS6: Salaries & Stipends	Compensation Intelligence	Full-time median midpoint ₹8.50L; internship median stipend ₹30,000.	Salary n = 59,871 (IQR: ₹2.75L); Stipend n = 40,129 (IQR: ₹14.9k).
PS7: Experience & Education	Qualifications & Entry Tiers	Strict separation: Interns = 0-1 yrs freshers; Full-time = 2-4 yrs mid-senior.	B.E./B.Tech required in 40.2%; MCA/MSc in 29.9%.
PS8: Career Recommendations	Practical Career Takeaways	Formulated 6 evidence-linked steps for skill acquisition and job search.	Directly derived from empirical dataset findings.

6. Actionable Career Recommendations for Students & Job Seekers

Grounding guidance in empirical evidence rather than speculation, our findings yield six actionable recommendations for students, early-career engineers, and academic mentors:

- **1. Benchmark Against the ₹30,000 Monthly Internship Standard:** Students with 0-1 years of experience should calibrate compensation expectations around the observed median stipend of ₹29,959 per month. AI/ML Engineering (₹30,056) and Cybersecurity (₹30,047) offer the highest student living allowances. Opportunities paying between ₹22,500 and ₹37,500 represent competitive market-rate entry positions.
- **2. Master High-Yield Cross-Cutting Skills First:** Python appears in nearly 40% of all tech postings, while Linux, Docker, and Git appear in ~20% of requisitions. Students should establish production-level fluency in Python and containerized Linux workflows before branching into niche frameworks. These skills provide multi-domain transferability across AI/ML, DevOps, Web Development, and Cloud.
- **3. Treat Technical Internships as the Mandatory Full-Time Gateway:** Because 100% of entry-level positions (0-1 years) in this dataset are structured as internships, students should not wait until graduation to apply for full-time vacancies. Securing a structured internship during academic semesters is the primary mechanism to bridge into permanent mid-senior positions (₹8.50 Lakhs median midpoint).
- **4. Avoid Premature Domain Arbitrage:** Full-time median midpoints exhibit less than 1% variance across categories (DevOps Core: ₹8.52L vs. Cybersecurity: ₹8.46L). Candidates should choose specializations based on intrinsic aptitude and portfolio strength rather than pursuing superficial salary premiums, as all six technical domains offer equivalent baseline compensation.
- **5. Capitalize on Geographic Salary Parity:** Employers maintain standardized national salary bands across Bangalore, Mumbai, Hyderabad, Pune, and Gurgaon (~₹8.50L median). Job seekers can expand their application scope nationally without incurring compensation penalties, selecting locations based on cost of living, mentorship quality, and living preferences.
- **6. Prepare for Substantial Salary Negotiation Spreads:** Employers post salary ranges with an average spread of ~₹8.0 Lakhs between minimum and maximum listings. Candidates who showcase verifiable production deployments, open-source contributions, and cloud certifications can effectively negotiate toward the 75th percentile (₹9.87 Lakhs+).

7. Methodological Disclosures & Quality Checklist

To ensure compliance with the evaluation criteria and enterprise data governance standards, the team confirms completion of the following mandatory disclosures:

- **Dataset Currency & Units:** All compensation values are confirmed in Indian Rupees (INR). Field names omit explicit annual/monthly periods; hence, full-time salaries and monthly stipends are reported descriptively without assuming take-home pay.
- **Skill Parsing Validation:** skills_extracted parsing was documented and verified; parsed list counts matched skill_count across 100% of records with zero mismatches.
- **Industry Representation Limitation:** Explicitly disclosed that 100% of records belong to Information Technology, precluding broad cross-industry comparisons.
- **No Predictive Overreach:** Analysis stops at descriptive EDA; no machine learning or clustering was built, and career_growth_index was reported without unverified forecast claims.
- **Reproducible Execution:** The entire workflow is codified in 'upto member 5.ipynb' and runs top-to-bottom cleanly without warnings or runtime exceptions.